Project Idea & Workflow

Title of project:-

Trinethra - Supervisor Feedback Analyzer

Main flow diagram:-

Supervisor Transcript

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prompt Engineering

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Local LLM (phi3:mini)

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structured JSON Output

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Validation layer

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Evidence verification

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Gap Analysis

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Frontend Dashboard

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Human Review

Architecture Overview

React Frontend

TranscriptInput + useAnalyze hook + API client + Dashboard

components: Scorecard, EvidenceList, Gap Analysis,

Follow up questions, Confidence Badge, KpiTags

| POST /api/analyze

Fast API Backend

routes.py → FeedbackAnalyzer.analyze( )

Jinja2 prompt Assembly

system_prompt.text + analysis_prompt.J2 (with schema)

OllamaClient (httpx async)

phi3:mini . local inference

JSON Extraction & Repair

strip fences → find { } bounds → fix trailing commas

pydantic schema validation

MVOutput (9 fields, typed)

Guardrails Layer

Evidence quote verification . KPI allowlist check

Deterministic confidence computation

Analyse Response

React Dashboard

Title :- Problems During Development

=> Large Prompts caused slow inference and unstable outputs

=> Local model generated malformed JSON responses

=> AI hallucinated unsupported observations

=> Low-RAM hardware limited larger models like Llama 3.2.

=> Nested schemas increased parsing failures

Small Reflection section

observation;
Smaller local models require simpler prompts and flatter response structures for reliable execution

Prompt Engineering

-> shorter prompts

-> JSON only

-> deterministic outputs

Validation

-> pydantic schema

-> retry handling

-> invalid JSON rejection

Evidence Layer

-> Evidence must exist in transcript

-> avoid hallucinated quotes

Human Review

-> AI assists reviewer

-> does not replace decisions

Goal:-

Convert unstructured supervisor feedback

Where I Disagree with AI

Title:- where I Disagreed with AI suggestions

Section 1

AI suggested:
- → larger prompts
- → complex schemas
- → more nested outputs

I changed
- → shorter prompts
- → flat JSON structures
- → lightweight validation pipeline

Reason
Local inference became unstable and slower with complex outputs

Section 2

AI suggested:
- → more features
- → additional scoring layers
- → advanced reasoning chains

I prioritized:
- → reliability
- → explainability
- → stable execution
- → human-review workflows

Section-3

AI suggested:

→ cloud-scale architecture

→ heavy pipelines

I intentionally kept:

→ lightweight MVP architecture

→ local-only inference

→ maintainable codebase

Engineering Decisions & Solutions

<u>Title</u>:- Solutions & Engineering Decisions

Prompt optimization

-> reduced prompt size
-> removed unnecessary instructions
-> used JSON-only prompting
-> added concise output format

Hallucination Control

-> evidence quote verification
-> transcript-based reasoning
-> confidence-aware outputs
-> strict schema validation

Model Decision

-> shifted from llama3.2 to phi3:mini
-> optimized for local inference
-> balanced reliability vs reasoning depth

JSON Reliability

-> flat JSON schema
-> retry prompts for Invalid responses
-> Custom JSON extraction logic

Frontend UX

→ modular dashboard cards

→ assesment coverage section

→ evidence visualization

→ confidence indicators

Bottom Tradeoff Note

Simpler architecture improved stability and reduced hallucination risk on low-resource hardware

Negative Prompting & Guardrails

Title:- Negative prompting & Guardrails

Negative Prompting used

→ Do not invent information

→ use only transcript evidence

→ Return valid JSON only

→ keep answers concise

→ Avoid unsupported assumptions.

Guardrails Implemented

→ evidence verification

→ schema verification

→ retry handling

→ confidence scoring

→ predefined KPI mapping

Human Review philosophy

The system is designed to support reviewers with structured analysis rather than replace professional judgement

Final Reflection Section

Key learning:

Building reliable AI systems requires balancing model capability, inference cost, explainability, and hallucination control.